

Project Summary

Overview

To enable urban informatics communities to effectively harness cyberinfrastructure (CI), artificial intelligence (AI), and real-time urban data for timely and data-driven urban management, this project establishes an international CyberTraining for AI-Enabled Intelligent Urban Informatics and Smart City Innovation (CITY-AI) network that broadens CI, AI, and streaming analytics skills across disciplines. To achieve this goal, we will: 1) establish the CI-enabled CITY-AI network of multiple academic disciplines, institutions, and professional organizations to leverage the expertise of communities in developing training and education materials on open platforms (e.g., I-GUIDE Platform) for preparing the next generation workforce; 2) develop a CITY-AI curriculum and training materials that will be openly accessible and implemented through webinars, workshops, and summer schools, which utilize CI and smart urban technologies for effective urban management practice; 3) introduce streaming data and CI to help participants leverage real-time and near real-time data sources available in modern cities; and 4) integrate the CITY-AI materials into existing curricula at Virginia Tech (VT) and University of Texas at Dallas (UTD). We aim to train over 1500 students, researchers, and educators through our partner networks. This project is led by researchers at VT and UTD who have been at the cutting edge of cyberGIS and urban informatics, partnering with an expert team at Purdue who operate production-scale national and campus advanced CI resources. This project will build and expand upon NSF-funded projects by the proposed team: Anvil - A National Composable Advanced Computational Resource for the Future of Science and Engineering (OAC-2005632); StreamCI: an AI-ready streaming data platform for a wide range of scientific applications (OAC-2513947); and HDR I-GUIDE (OAC-2118329).

Keywords: Urban Informatics Workforce, CI-Enabled Urban Analytics, Streaming Data, GeoAI Smart City Management, Geospatial Analytics, Level of Students

Intellectual Merit

The wide availability of urban streaming data (e.g., sensors, GPS) and AI provide an unprecedented opportunity to harness real-time data and support timely decision-making for urban environments like transportation and emergency response. Current and future generations of the urban management and planning workforce require harnessing CI resources, large-scale geospatial data and AI to seize this opportunity. The intellectual merit of this project includes: 1) a CI-enabled geo-computational urban informatics research network across disciplines including GIS, computational science, engineering and urban planning; 2) a CI-focused urban informatics curriculum consisting of summer schools, workshop sessions, and online webinars; 3) modular training materials that will support existing NSF efforts, adopted at VT and UTD; 4) openly accessible urban informatics training materials developed using FAIR (Findability, Accessibility, Interoperability, and Reusability) principles leveraging existing NSF-funded CI capabilities; and 5) cutting-edge case-studies to demonstrate the capabilities to participants.

Broader Impacts

CITY-AI will increase adoption of CI for urban informatics research communities and enhance workforce development among disciplines including urban planning, geography, computer science, public health, economics, political, and social sciences. The broader impacts of this cybertraining workshop are multifaceted. First, we aim to provide urban streaming data, online training material, workshops and CI resources to train a next-generation workforce that can meet the growing demand of smart city management with the expertise of AI and CI. Second, the curriculum, workshops and network will build connections between the urban sciences and the computational science and engineering (CSE) communities, specifically familiarizing them with the NSF ACCESS and NAIRR projects. Third, our course materials will be openly available and developed as stand-alone courses that are accessible to the general public and can be used by educators in classrooms of all levels and fields. The project will be scaled up and sustained through our core organizations by partnering with an NSF-funded center (I-GUIDE), urban streaming data producers (Purdue Rosen Center for Advanced Computing, VT Center for Geospatial Information Technology), and professional organizations (University Consortium for Geographic Information Science, American Association of Geographers).

Project Description

1. Challenges for Research Workforce Development

Cities now produce large and streaming geospatial datasets from sensor networks, GPS, remote sensing observations, and crowdsourced platforms [1], [2]. These data capture rapid changes in transportation systems, urban environment, infrastructure, and human activity [3], [4]. The amount of effort collecting urban streaming data has grown significantly, for example, the city of Chicago has launched the Array of Things (AoT) to capture streaming urban environment data as the fitness tracker of the urban area [5], the city of Los Angeles developed Advanced Transportation System and Coordination (ATSAC) for real-time centralized traffic management [6]. The ability to analyze high-volume and high-velocity data has become critical to research and practice in urban analytics and management. At the same time, advances in Artificial Intelligence (AI) and cyberinfrastructure (CI) have created new opportunities for real-time and (near) real-time modeling, scalable geospatial computation, and data-driven decision support [1], [7], [8]. These capabilities are increasingly used in applications such as mobility prediction [9], heat exposure assessment [10], [11], infrastructure monitoring [12], and emergency response [13], [14]. As urban informatics continues to rapidly evolve, a significant disconnect exists between industry/government practices and academic training.

Yet urban informatics communities play an increasingly important role in supporting real-time decision-making, these activities require the ability to process large and dynamic urban datasets, which many students and early-career researchers lack. Although many tools and urban management systems used in real-world urban management are initially prototyped in university research, academic curricula, particularly at the undergraduate level, often lag behind practice. As a result, students in urban-focused disciplines frequently graduate without formal training in large-scale, real-time data analysis techniques, CI, and AI-enabled workflows used in contemporary urban management and urban informatics research. Urban planning, geography, and other urban-related programs emphasize qualitative analysis and policy-oriented approaches, with limited integration of data-intensive methods, scalable computation, or AI-based analytics, which will be essential in the future urban management and research [15], [16]. Consequently, the workforce responsible for implementing and managing real world urban systems is often unable to fully benefit from formal academic preparation in the tools and computational approaches they are expected to use in practice, demanding a comprehensive curriculum and training materials focusing on urban informatics. Addressing these gaps requires coordinated workforce development that connects urban science with computational and AI methods.

1.1. Barriers to CI access in Urban Informatics Communities

Despite recent NSF investments that have expanded access to national CI resources, many urban informatics communities remain underrepresented among active users of advanced computing systems [17]. Table 1 summarizes ACCESS usage patterns, which show that geosciences, engineering, and social science fields - areas central to urban informatics - have comparatively low numbers of users and low percentages of resource utilization relative to mathematical, physical, and biological sciences [18]. Several factors contribute to this underuse: limited awareness of available CI platforms, uncertainty about how these systems can support urban research and management workflows, expected high learning curves for CI resources, and a general lack of CI training across urban-focused programs [19]. As a result, many researchers may not fully recognize the need for CI in urban analytics or understand how these resources can enhance their ability to process large datasets or support real-time decision-making in everyday urban operations. Many urban planning and geography departments rely primarily on desktop GIS labs that can't support scalable, data-intensive, or streaming-data workflows. Urban informatics draws on geography, computer science, engineering, transportation studies, environmental science, and social sciences, yet many of these fields depend on small-scale analytical tools that are insufficient for processing large-scale urban datasets, conducting city-scale simulations, or training modern deep learning models [7].

Table 1. NSF ACCESS Utilization from 01/01/2020 to 12/31/2025		
ACCESS Usage by NSF Directorate	ACCESS Credit (million)	Percentage Utilization
Mathematical & Physical Sciences	8,246	55%
Biological Science	2,815	18.8%
Engineering	1,669	11.1%
Geosciences	1,201	8.0%
Computer & Information Science	993	6.6%
Social, Behavior & Economic Sciences	34.7	0.23%

Low ACCESS utilization in urban informatics communities reflects both training limitations and a lack of integration of applications between existing urban management workflows and CI capabilities [20]. As cities increasingly depend on real-time data analysis for transportation operations, hazard response, and climate adaptation, the absence of a computationally trained workforce poses growing challenges for scientific progress and operational effectiveness [21]–[23]. The CyberTraining for AI-Enabled Intelligent Urban Informatics and Smart City Innovation (CITY-AI) program is designed to address these barriers by providing domain-relevant CI training, openly accessible learning materials, and a coordinated educational network that enables researchers and practitioners to use CI-enabled, AI-driven urban analytics.

2. Solicitation Goals Targeted

This project will establish the CyberTraining for AI-Enabled Intelligent Urban Informatics and Smart City Innovation (CITY-AI) network, bringing together urban informatics research communities across disciplines, careers, and levels of expertise to broaden their computational and cyberinfrastructure skills and help them harness real-time data from modern data-rich cities.

2.1 Broadening Adoption of Advanced CI

This project targets researchers across many domains focused on urban informatics and smart city innovation to ensure broad adoption of CI tools, methods, and resources by the research community in order to catalyze major research advances and to enhance researchers' abilities to lead the development of new CI. Our training program will provide researchers with the critical computational skills to utilize CI and demonstrate how CI can be applied to their research.

We aim to reach more than 1,500 participants through a variety of activities. This includes in-person workshops will be run at conferences organized by the American Association of Geographers (AAG), University Consortium for Geographic Information Science, the Association of Collegiate Schools of Planning (ACSP), and the Practice and Experience in Advanced Research Computing (PEARC) conference; webinars showcasing related research that is leveraging CI; and a summer school that will deliver our curriculum over the course of a week along with a mentored hands-on research experience. In addition to the participants in our training and webinars, we will establish a CITY-AI network that shares related training and opportunities to promote continued education.

Table 2. Training activities and estimated participation.

CITY-AI Event	Year	Participants (per session)	Total Participants
Webinars (4 webinars/year for 3 years)	1-3	50	600
AAG Annual Meeting 2027-29 (2 sessions/year, 3 years)	1-3	50	300
PEARC Conference (2 sessions/year, 1 year)	2	50	100
UCGIS Symposium (2 sessions/year, 1 year)	3	50	100
ACSP Annual Conference (2 sessions/year, 1 year)	3	50	100

Summer Schools (Year 3)	3	20	20
Online Curriculum Modules	1-3		300
Total			1,520

2.2 Integration of CI into Curriculum

This proposal also supports the second solicitation goal, integrate core literacy and discipline-appropriate advanced skills in advanced CI as well as computational and data-driven methods for advancing fundamental research, into the Nation's undergraduate and graduate educational curriculum/instructional materials, by developing 15 lessons, 12 webinars, 12 workshops, and a summer school based on the following specific objectives:

- **Objective 1:** Create 15 2-hour CI-enabled urban informatics modules covering HPC training and critical skills for harnessing real-time and near real-time data streams. Each lesson will consist of a 1-hour lecture followed by a 1-hour hands-on lab. The training material will be continually updated throughout the project based on feedback from students and collaborators.
- **Objective 2:** Distribute our modules through CITY-AI activities including 12 webinars, a 5-day long summer schools, and workshop sessions at conferences including the AAG Annual Meeting, UCGIS Symposium, ACSP Annual Conference, and the PEARC conference.
- **Objective 3:** Create a repository of CI-oriented open educational resources (OERs) that can be used by educators, students, and professionals to enhance their learning experience. This will include recordings of workshops, webinars, and course materials with lesson plans and hands-on activities.
- **Objective 4:** Create and promote a CITY-AI network of training participants, students, educators, and partners to gather feedback, distribute our curriculum, and promote continuing training. This includes networking events at conferences and creating a newsletter to distribute opportunities throughout the community.

The content of the courses aligns with the institutional priorities and will be integrated into undergraduate and graduate coursework at UTD and VT. The training material will be openly available through our project website and nationally recognized CI Platforms such as the I-GUIDE Platform (#2118329) [24], and cybertraining platform called CyberFaCES run by Purdue Rosen Center For Advanced Computing (RCAC). We might be able to link to your course to help attract potential learners. CITY-AI events and training materials will also be shared with research communities via institutional and academic mailing lists.

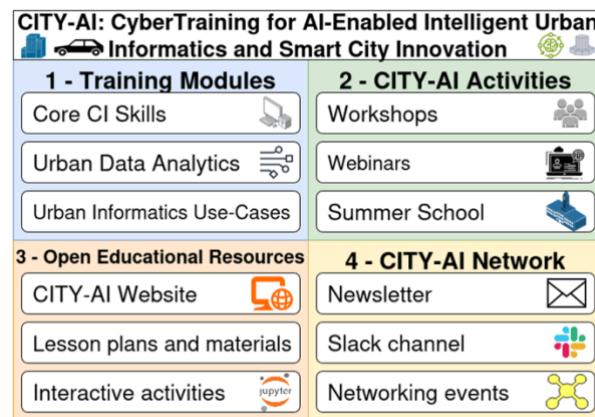


Figure 1. Overview of CITY-AI Objectives

2.2.1 Training Modules (Objective 1)

Urban studies curricula often emphasize qualitative understanding and policy-oriented analysis, leaving a critical gap in quantitative, technology-driven skills necessary for modern urban management [25]. With the rapid advancement of artificial intelligence, cyberinfrastructure, and the increasing availability of real-time urban data, there is a growing need for precision, effectiveness, real-time and data-driven decision-making in urban planning and governance [26], [27]. CITY-AI aims to address this gap by developing an innovative curriculum that integrates urban complexity modeling, AI-enabled analytics, and scalable computing tools. Our curriculum will offer transdisciplinary training and hands-on practice with real-world datasets, empowering learners to contribute to responsive, efficient, and resilient urban systems.

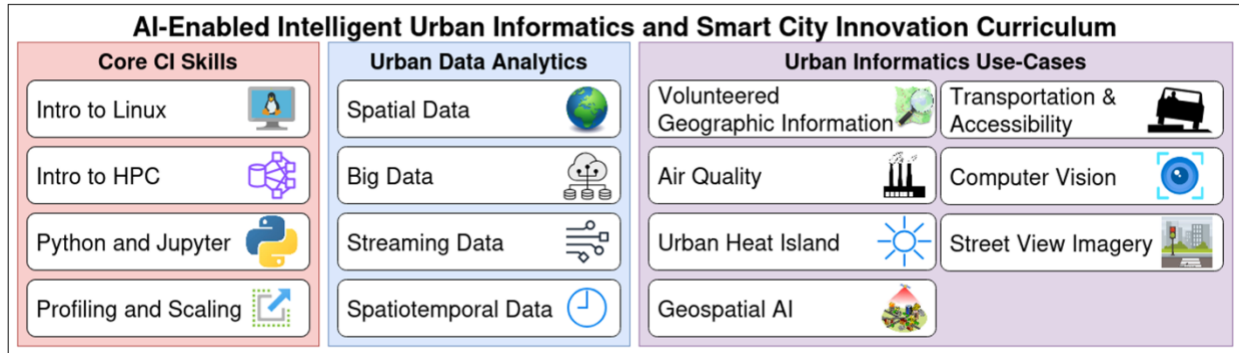


Figure 2. An overview of CITY-AI Curriculum.

Our collaboration network members will work with us to identify learning goals, develop course content, refine the materials, and amplify CITY-AI materials throughout their networks. Preliminary training modules are given in Table 3 listing the overall theme of the module, the topic, the lead developer of that module, the difficulty level (Beginner [Beg], Intermediate [Int], or Advanced [Adv]), and overall learning goals for that module.

- **Core CI Skills (Song Lead):** Familiarize participants with CI and HPC to lower technical barriers for urban informatics researchers and practitioners. These modules will provide critical technical skills for leveraging CI and provide hands-on experience.
- **Urban Data Analytics (Michels Lead):** Builds the core spatiotemporal analytics skills to leverage real-time data streams to address urban problems. Participants will learn to work with common spatial datasets (e.g., census, remote sensing) and common spatio-temporal (ST) analysis techniques. We will also teach participants the challenges of working with big data, collecting data from streams, and the basics of analyzing, modeling, and visualizing ST data.
- **Urban Informatics Use-Cases (Lyu Lead):** Introduces common datasets and problems that benefit from real-time urban analytics. Building on the Core CI Skills and Urban Data Analytics modules, these modules will give participants hands-on experience with real-world datasets and problems including air quality, urban heat, transportation, and street view imagery.
- **Urban Informatics in Action (UIA) (Lyu Lead):** The CITY-AI training modules will be supplemented with UIA webinars that integrate the concepts from the three previous themes and illustrate real-world problem solving with real-time data. Example talks are in Table 4.

Table 3. Overview of the CITY-AI training modules.

Theme	Module Topic	Lead	Level	Learning Goals
Core CI Skills	Intro to Linux	Song Michels	Beg	Introduces the terminal, file management, I/O, and concepts required to use HPC systems.
	Intro to HPC	Song	Beg	Learn to use NSF ACCESS resources including Anvil. Basics of OpenOnDemand and batch processing.
	Python and JupyterLab	Song Michels	Int	Introduce the Python programming language and JupyterLab interface; work with core packages like numpy, scipy, and matplotlib.
	Profiling and Scaling	Song	Adv	Introduces computational complexity, basic approaches to estimate computational needs, and strategies to scale software.
Urban Data Analytics	Spatial Data	Michels	Beg	Learn to work with geospatial data: core concepts, data types, and mapping.

	Big Data	Michels	Int	Addresses the challenges and opportunities of working with big data.
	Streaming Data	Lyu	Int	Fundamentals of obtaining, cleaning, and processing streaming data sources with examples of real-time urban data streams.
	Spatiotemporal (ST) Data	Lyu	Adv	Data analysis and visualization techniques for ST data including clustering and statistics.
Urban Informatics Use-Cases	Volunteered Geographic Information (VGI)	Michels	Int	Introduce VGI and learn how to work with common VGI data sources like OpenStreetMap (OSM) to analyze and model urban environments.
	Air Quality	Lyu	Int	Learn to work with real-time temperature sensors and spatio-temporal data processing.
	Urban Heat Island (UHI)	Lyu	Adv	Learn to combine sensor data with other geospatial data to explore the UHI effect.
	Geospatial AI	Lyu	Int	Introduce AI and understand how spatial and time can be included in AI models to solve urban problems.
	Transportation & Accessibility [28], [29]	Michels	Adv	Learn to use real-time data sources like real-time GTFS to model transportation and spatial accessibility problems.
	Computer Vision (CV)	Lyu Michels	Adv	Learn to work with imagery data, pre-trained CV models, and basics of training/fine-tuning.
	Street View Imagery (SVI)	Lyu	Adv	Learn to process SVI and apply CV models for object detection and image segmentation.
	Remote Sensing	Lyu	Adv	Learn to process urban remote sensing data and apply machine learning models for image classification and segmentation.

Table 4: Example presentations for the Urban Informatics in Action webinars

CITY-AI Urban Informatics in Action Webinars
Extractive versus Generative Language Models for Political Conflict Text Classification (Brandt, UTD, USA)
Exploring the inequality of the particulate matter exposure level of public bicycle scheme users using an agent-based model (Kang, Kyung Hee University, SK)
A Video Machine Learning Framework for Spatiotemporal Analysis of Complex Urban Dynamics (Lyu, VT, USA)
Centennial Reflections: A Century of Shaping Tomorrow's Cities (Song, UNC, USA)
Reproducibility and Replicability in the Context of the Contested Identities of Geography (Sui, VT, USA)
I-GUIDE Platform for Geospatial Data-intensive Convergence Research and Education (Wang, UIUC, USA)
Enhancing Address-Level Broadband Mapping to Close Virginia's Digital Divide: A State-University Collaboration (Brandon, VT, USA)
High-performance and cloud computing on Anvil and ACCESS resources and container basics (Chris

Thompson, Purdue University, USA)
Streaming data cyberinfrastructure StreamCI (Jaewoo Shin, Purdue University, USA)
Multimodal Urban Retrofitting Detection: A Vision Transformer approach using Street View Imagery and Demographic Data (Raj Bhattarai, VT, USA)
Leveraging temporal changes of spatial accessibility measurements for better policy implications: a case study of electric vehicle (EV) charging stations in Seoul, South Korea (Jinwoo Park, Kyung Hee University, SK)
Revitalizing Cities: The 5R Framework Approach to Urban Retrofitting and Big Data Insights (Xinlin Ma, VT, USA)

CI Resources and Training Environment

Participants will learn to use NSF ACCESS resources and the I-GUIDE Platform (#2118329) for urban informatics and real-time data processing. Co-PI Song’s NSF Anvil (#2005632) is part of NSF ACCESS and the primary HPC used by participants. Anvil is a national-scale computing resource featuring a large CPU based system and advanced accelerators such A100 and H100 GPUs, a Kubernetes-based composable system, as well as a multi-tier high-performance data storage system. The Anvil ecosystem includes a comprehensive software stack, including 800+ optimized containers of science applications, modern services such as AnvilGPT and Notebook Service, and interactive access interfaces. The Purdue Anvil team has supported more than 14,000 users, including 6000 undergraduate students learning data science and computing, to date. PI Song will guide students through account creation for NSF ACCESS and provide them access to NSF Anvil for training. PI Lyu and Co-PI Michels will train students to use the I-GUIDE Platform [24] and its associated JupyterHub.

Our emphasis on online learning environments and OERs will ensure that our training modules scale to larger audiences and can be used outside of our direct activities. Modules will be demoed and refined through use in classrooms at VT and UTD, our in-person workshops at conferences, and summer school. These refined materials will then be hosted as OERs that include lesson plans and hands-on activities through Jupyter notebooks [30]—an open-source, interactive, and approachable Graphical User Interface for displaying Markdown and executing code—which can be executed on the I-GUIDE Platform’s JupyterHub. The I-GUIDE JupyterHub is built on a framework that provides version-controlled software environments, ensuring our materials will be executable even as the system is updated [31]. These materials will be hosted on a CITY-AI website, hosted through Github pages, allowing for collaborative curriculum design and providing both instructors and participants built-in forums to provide feedback. The website will be accompanied by a Slack channel open to the CITY-AI network which can be used to share opportunities, provide feedback, and foster community.

Training Module Development Goals

We will develop a set of guidelines that specify the learning outcomes, structure, and knowledge used to develop each module. This document will be developed at the beginning of the project and ensure that our modules are self-contained while still fulfilling the overall goals of the CITY-AI training curriculum. PI Lyu and Co-PI Michels will engage free resources at the VT Center for Excellence in Teaching and Learning and the UTD Center for Teaching and Learning to conform to teaching best practices which will be shared with the CITY-AI team. We will also develop and administer evaluation forms for participants of CITY-AI Activities to fill out after each training, using this feedback to continually improve and refine our modules. CITY-AI modules will also provide connections to external education and training resources such as the UCGIS Body of Knowledge [32], the I-GUIDE Convergence Curriculum [33], and the Hour of CI project [34].

2.2.2 CITY-AI Activities (Objective 2)

The CITY-AI training consists of a mix of education activities to promote collaboration, education, and workforce development in real-time urban analytics:

- **Workshops:** We will organize workshop sessions at the AAG annual meeting (Years 1-3), UCGIS Symposium (Years 2-3), and the ACSP Annual Conference (Years 2-3). Workshop content will serve as the basis for the training modules in Table 3, constructed with feedback from our collaborators. The AAG and UCGIS conferences were selected as large multi-disciplinary conferences focused on spatial and geographical sciences and the ACSP Annual Conference was selected as a more public policy and professional focused venue. The first session at each conference will focus on CI skills, introducing the CI platforms we are using (NSF ACCESS, Anvil, and I-GUIDE Platform), and the Jupyter interface. Our second session will provide an application-driven introduction to streaming data and machine learning. Both sessions will heavily focus on hands-on activities, introduce the CITY-AI network and resources, and discuss how participants can leverage our resources for continued learning. We plan to host at least 50 participants per session at each of these sessions and have budgeted to provide travel awards for a few students for each conference.
- **Webinars:** Our Urban Informatics in Action (UIA) webinars will focus on novel, real-world applications of AI-Enabled Intelligent Urban Informatics and Smart City Innovation. Our goal with these webinars is to reach audiences interested in the topics (e.g., digital twins, smart cities) and demonstrate how CI and AI are being used to overcome challenges, piquing audience interest in the CITY-AI curriculum and network. Talks will come from our current collaborators and our growing network of educators, researchers, and participants. We plan to organize three webinars per semester to avoid the busy early and end of semester timeframes and we estimate at least 50 participants per webinar.
- **Summer School:** The summer school will be a week-long learning experience coupled with a research problem to tackle, organized at UTD during year 3. We will utilize training materials and OERs developed for workshops, following the curriculum described in Table 3. We plan to host 20 students for a week and have budgeted to pay all expenses for the students (e.g., travel, room, board) to ensure broad participation. Participants will be recruited through academic mailing lists, conference workshops, and the CITY-AI Network (see 4. Recruitment and Evaluation).

2.2.3 CITY-AI Open Educational Resources (OERs) (Objective 3)

Our effort in creating the CITY-AI Curriculum (Obj 1) and delivery of the curriculum (Obj 2) will provide sustained benefits to the community by providing the curriculum as open educational resources (OERs). Hosted on the CITY-AI website and executable on the I-GUIDE Platform, our curriculum will provide learning materials practitioners, educators, and students who want to expand their CI knowledge. Each OER will include lesson plans, slides, an interactive activity like a Jupyter notebook, and recommended readings. In addition to our OERs, our website will host recordings of our webinars and workshops, links to our Slack community to engage with the CITY-AI community and provide feedback, and a calendar of upcoming activities.

2.2.4 CITY-AI Network (Objective 4)

We will create and promote a CITY-AI network of training participants, students, educators, and partners to gather feedback, distribute our curriculum, and promote continuing training. This network will be grown through CITY-AI activities and our newsletter, inviting participation from academia, research centers, non-profits, governmental agencies, industry, and computing centers hoping to promote their opportunities, share their work, or utilize our curriculum. Growth in the CITY-AI network and in the professional networks of CITY-AI participants will be stimulated by organizing networking events at conferences and the creation of a Slack channel. This network will help us to recruit participants, gather feedback on our curriculum, and increase engagement with our OERs.

CITY-AI Leadership

- Fangzheng Lyu (PI): Assistant Professor of Geographic Information Science and Director of GeoCompute Research group at the Department of Geography in Virginia Tech. He specializes in urban informatics [10], [11], [35]–[37], spatial AI [35], [38]–[40] and geo-computing [41]–[43]. He is the Chair of AAG CyberInfrastructure Specialty Group, membership committee member of UCGIS.
- Alexander Michels (Co-PI): Assistant Professor of Geospatial Information Science in the School of Economic, Political, and Policy Sciences at UTD has expertise in cyberinfrastructure (CI) and cyberGIS [24], [29], [31], [44], [45], big data [28], [46], and health geography [47]–[49]. He is a Director of the AAG CyberInfrastructure Specialty Group, serving on the board since 2021. He has led multiple conference workshops on topics including CI and GIS, organized and chaired symposiums, and served as an instructor at the 2025 I-GUIDE Summer School. In 2026, he was selected as a UCGIS I-GUIDE Community Champion to develop open science materials focused on community outreach.
- Carol Song (Co-PI): is Chief Scientist for the Purdue Rosen Center for Advanced Computing (RCAC) and PI for NSF OAC-2005632, Category I: Anvil - A National Composable Advanced Computational Resource for the Future of Science and Engineering, \$31,497,100, 10/2020-9/2027. Anvil expands the national advanced computing cyberinfrastructure (CI) with a forward-looking composable computing resource that integrates a large-capacity high-performance computing (HPC) system, advanced GPUs and a Kubernetes-based on-prem cloud subsystem, and a comprehensive ecosystem of software, access interfaces, programming environments, and composable services to form a seamless environment capable of supporting a wide range of current and future science and engineering (S&E) applications. Anvil lowers barriers to HPC access with user-friendly interfaces and supports diverse S&E applications through Kubernetes-based composability and cloud-bursting. Its training programs and partnerships build next-generation CI and computing expertise.

Unpaid U.S. and International Collaborators

- Dr. Daniel Sui: Senior Vice President, Chief Research & Innovation Officer, Professor of Geography and Public & International Affairs at Virginia Tech with expertise in GIScience and Volunteered Geographic Information.
- Dr. Shaowen Wang: Associate Dean for the College of Liberal Arts & Sciences, Professor of Department of Geography & Geographic Information Science and Siebel School of Computing at University of Illinois Urbana-Champaign, Director of NSF I-GUIDE. His research interests primarily include GIS and cyberGIS.
- Dr. Kathleen Stewart: President of the University Consortium of Geographic Information Science and Professor, Department of Geographical Sciences at the University of Maryland.
- Dr. Yan Song: Professor at the Department of City and Regional Planning, Director of the Center for Urban and Regional Studies at the University of North Carolina at Chapel Hill. Editor-in-chief of Journal of the American Planning Association (JAPA). Her expertise is in low carbon and green cities, plan evaluation, land use development and regulations.
- Brandon Herndon: Director of the Center of Geospatial Information Technology (CGIT) at Virginia Tech.
- Dr. Patrick Brandt: Professor of Political Science at the University of Texas at Dallas, Faculty Affiliate of the Texas Artificial Intelligence Institute, and Research Faculty of the Global Counterterrorism and Threat Intelligence Research Institute has research experience in machine learning, large language models, and streaming data.
- Dr. Jeon-Young Kang: Assistant Professor of Geography at Kyung Hee University in South Korea works in agent-based modeling, geospatial big data, and geospatial intelligence.

3. Scalability & Sustainability

The CITY-AI project is designed to scale and sustain by leveraging the project’s partnership strategy, modular curriculum design, and the existing national and international collaboration networks of the project team. During the project period, CITY-AI will train over 1,500 students, researchers, and

educators through webinars, conference workshops, and summer schools. The project will continue to scale by recruiting participants through professional organizations and CI platforms and by offering training materials in both synchronous and asynchronous formats to reach broader audiences beyond the core events and offer online training materials on the existing CI platform (I-GUIDE platform). Additionally, CITY-AI will scale participation by leveraging established networks in urban informatics, geospatial science, and cyberinfrastructure. These include professional societies (e.g., UCGIS, AAG), CI platforms (e.g., NSF ACCESS, I-GUIDE), and partner institutions. Conference-based workshops (e.g., AAG Annual Meeting, UCGIS Symposium) will provide access to large, interdisciplinary research communities, while open webinars and summer schools will extend participation nationally to train the next generation workforce. The project will also engage K–12 and undergraduate audiences through publicly accessible online materials and recorded training sessions, enabling downstream use by educators and outreach programs.

To support sustainability, all CITY-AI training modules will be developed as modular, reusable, and openly accessible resources available on an online NSF-funded platform. Training materials will be primarily implemented using Jupyter Notebooks and containerized environments with Docker image to ensure portability across CI platforms and institutional infrastructures. Materials will be hosted through the CITY-AI project website and shared via nationally recognized CI education platforms, including the I-GUIDE Platform and the NSF ACCESS Knowledge Base. This approach ensures that materials remain available beyond the project period and can be continuously reused and adapted by other institutions and training programs. CITY-AI will further sustain impact through partnerships with CI providers and data producers. The project builds on prior and ongoing NSF-funded efforts of the team, including Anvil (OAC-2005632), StreamCI (OAC-2513947), and HDR I-GUIDE (OAC-2118329). These platforms provide production-scale CI resources, training infrastructure, and long-term institutional support that will allow CITY-AI materials and workflows to remain operational and relevant as CI technologies evolve.

3.1 Alignment with Institutional Priorities

CITY-AI is strongly aligned with the educational and research priorities of the participating institutions. At VT, the project aligns with ongoing efforts in the University's priority in AI for science and the Department of Geography to expand training in GIS, geospatial data science, and urban analytics at both undergraduate and graduate levels. CITY-AI modules will be integrated into existing courses in VT (i.e., GEOG4424 Geospatial Data Science, GEOG5254 Spatial Data Science, GEOG2984 Geospatial Tools), supporting long-term curriculum enhancement leveraging the materials developed by CITY-AI.

At the University of Texas at Dallas, CITY-AI supports the educational and research focuses on AI and public policy. UTD's School of Economic, Political, and Policy Sciences (EPPS) is hoping to grow its program and course offerings at the intersection of AI and public policy, including public health, geospatial data science, and sustainability science. UTD is also building a Texas Artificial Intelligence Institute. Materials will also support ongoing courses at the undergraduate and graduate level (e.g., GISC 4381/6321 Spatial Data Science, CRIM 6332/GISC 6331 GIS Applications in Criminology). This work also aligns with institutional research focuses including the Texas Artificial Intelligence Research Institute.

Purdue University's Rosen Center for Advanced Computing provides a strong institutional foundation for sustaining CITY-AI through access to CI systems, user support expertise, and established CI training programs. By embedding CITY-AI modules within existing CI training ecosystems and academic curricula at partner institutions, the project ensures that training activities continue beyond the funded period and remain responsive to evolving workforce needs in urban informatics.

4. Recruitment and Evaluation

The CITY-AI team will develop and maintain a project website that provides centralized access to recruitment materials, activity schedules, syllabi, instructor information, and training resources. Each year, application and registration materials will be announced through the CITY-AI website, professional society mailing lists (e.g., AAG, UCGIS, VT, UTD), CI platform communication channels, and partner institution networks. Recruitment strategies will vary by activity type. Webinar participants will be recruited through open registration and professional mailing lists. Conference workshops will recruit participants primarily through conference networks and session announcements, with a mix of in-person and online participation options when feasible. Summer schools will use a competitive application process, collecting basic applicant information such as academic status, institutional affiliation, prior experience, and participation preferences through an online form. Travel support will be provided for selected in-person activities (workshops and summer school), with preference given to students from institutions with limited CI resources, and early-career students/researchers. Recruitment materials will explicitly encourage participation from urban planning, geography, social science, and interdisciplinary programs that have historically had limited engagement with CI but represent a major workforce of urban management.

The CITY-AI project will implement a comprehensive evaluation plan to assess program effectiveness, guide continuous improvement, and measure workforce development outcomes. The evaluation will include both formative and summative components and will focus on the following areas:

- Course Design and Implementation: Alignment of learning objectives with CI and urban informatics workforce needs; clarity, accessibility, and usability of training materials.
- Effectiveness: Coordination among project personnel; successful delivery of planned activities; achievement of participation targets.
- Impacts: Changes in participant knowledge, skills, and confidence in using CI and AI for urban analytics; adoption of CITY-AI materials in academic courses and research workflows.
- Sustainability: Continued use and dissemination of training materials; integration into institutional curricula; reuse of materials by external educators and programs.

Evaluation data will be collected through participant surveys, attendance records, activity logs, and usage metrics for training materials. Surveys will be administered following major activities to assess satisfaction, learning outcomes, and perceived barriers. Results will be reviewed regularly by the project leadership team and used to refine curriculum content, delivery formats, and recruitment strategies. Annual evaluation summaries will be incorporated into project reports to NSF. The team will also disseminate evaluation findings and best practices through presentations at relevant conferences (e.g., AAG, UCGIS) to support broader adoption of effective CyberTraining approaches in urban informatics.

Table 5. The illustration of evaluation metrics for the CITY-AI program.

Program Activities	Evaluation Metrics	Evaluation Method
Participant Recruitment	1. Number of students who applied (accepted) for the program. 2. Participants' demographics (academic status, institution type).	1. Survey for registration. 2. Recruitment and attendance records.
Course Design & Implementation	1. Number of students who attended the class. 2. Number of questions raised during	1. Attendance report. 2. Project website, I-GUIDE platform and Team channel data.

	and after the class. 3. Completeness of the class material regarding the timeline. 4. Integration of the CITY-AI material into the institutional curriculum.	3. Evaluation survey to track the satisfaction with the material. 4. The number of modules available to use.
Impacts and Sustainability	1. Number of publications directly or indirectly generated by the training module. 2. Number of data record and course material downloads/used.	1. Track participants' Google Scholar profiles. 2. Track the I-GUIDE platform and project websites on the number of views and downloads.
Training Quality	1. Software and hardware accessibility 2. Instructor evaluation 3. Students, instructors, and collaborators' satisfaction	1. Evaluation survey results.

5. Collective Impact Strategy

We will develop Working Groups of researchers and educators from national centers, urban informatics data producers, educational organizations, industry, and CI resources producers. Our Working Groups will coordinate with a set of Supporting Organizations to scale and sustain our work.

5.1 Working Groups

We developed four Working Groups with focuses on curriculum development, education outreach, CI resources coordination, and integration & evaluation. Our PIs will contribute to the working groups. Detailed information about each team member's responsibilities is outlined in the management plan.

- Curriculum Development Working Group (Lyu Lead): This group is responsible for the development and refinement of curriculum and online training materials. The group will meet regularly to discuss the course content and refinement and design the training structure for the summer school and workshop activities.
- Education Outreach Working Group (Michels Lead): This group will prepare and host workshops, webinars and summer schools. The group will also coordinate with other groups and supporting organizations for hosting CITY-AI events.
- CI Resources Working Group (Song Lead): This group will implement the CI computing environment for the CITY-AI courses. The group will also discuss and resolve all the technical problems faced during the training, e.g., login to CI resources, remote access, code distribution, and other technical details.
- Integration & Evaluation Working Group (Lyu Lead): This group will design and implement strategies of integrating CI and GIS into urban informatics domain and conduct an evaluation plan to provide formative feedback for guiding program planning and implementation.

5.2 Supporting Organizations

Our team will work with the following to help the team to scale up and sustain the program.

- AAG (PI Lyu as a Chair for the Cyberinfrastructure Specialty group; Co-PI Michels as Director for the Cyberinfrastructure Specialty group) has 10,000 registered members. The annual meeting has 9,000+ attendees and thousands of presentations in paper, poster, and panel sessions covering all aspects of geography.
- UCGIS (Stewart as President, see letter; PI Lyu as member of Membership Committee): creates and supports communities of practice for GIScience research, education, and policy endeavors in higher education. UCGIS symposium brings GIScience students, researchers, and educators together every

summer to discuss key topics in GIScience education and research. UCGIS is a non-profit organization that has 70 partner institutions across the globe.

- Rosen Center for Advanced Computing (#2005632, co-PI Song as Chief Scientist) provides advanced computational resources and services to research and education.
- NSF-funded Anvil (co-PI Song as PI), consisting of 1000 128-core computing nodes based on the next-generation AMD Epyc “Milan” architecture that can deliver a total peak performance of 5.3 Petaflops, represents a national advanced computational resource for science and engineering research and education.
- NSF I-GUIDE (#2118329, Wang as PI, see letter) enables discovery and innovation for tackling fundamental scientific and societal challenges by harnessing the vast, diverse, and ever-growing corpus of geospatial data. It also has the I-GUIDE Platform that enables users to discover resources and execute interactive workflows implemented as Jupyter notebooks.
- Texas Artificial Intelligence Research Institute (Brandt as Faculty Affiliate, see letter) brings together an interdisciplinary team of experts to pursue groundbreaking research across the full spectrum of AI—from foundational theory to transformative applications advancing the sciences and human health.
- University of North Carolina Center for Urban and Regional Studies (Y. Song as Director, see letter) seeks to generate new knowledge of urban and regional processes and problems and ultimately to improve living conditions in our communities, representing the urban planning community that we aim to involve in this project.
- VT Center for Geospatial Information Technology (Herndon as Director, see letter) collaborates across research, education, and outreach with a transdisciplinary approach, addressing complex problems with geospatial science.

6. Fostering Community

CITY-AI represents a cross-disciplinary effort spanning urban informatics, geography, and cyberinfrastructure (CI). The project targets three primary communities to expand its impact. First, geography and GIScience communities will be engaged through workshops and sessions at major venues such as the Annual Meeting of the AAG and the UCGIS Symposium, leveraging established professional networks to promote CI- and AI-enabled urban analytics. Second, urban planning communities will be reached through workshops at conferences such as the Association of Collegiate Schools of Planning (ACSP) Annual Conference and through collaborations with university-based urban planning and policy centers (e.g., UNC, Virginia Tech), increasing visibility and adoption of CI-enabled methods in planning education and practice. Third, CI and computer science communities will be engaged through participation in CI-focused meetings (e.g., PEARC), highlighting urban informatics as a high-impact application domain for AI- and CI-enabled science.

Community building is further reinforced through openly accessible training materials, hands-on labs delivered via NSF-funded platforms (e.g., I-GUIDE), and persistent online engagement through the project website and shared repositories. Partnerships with established professional organizations (e.g., AAG, UCGIS) and NSF CI platforms (e.g., ACCESS, I-GUIDE) ensure credibility, scalability, and long-term sustainability. Continuous feedback from participants will guide refinement of activities and resources, enabling CITY-AI to evolve with community needs and advances in CI-enabled urban analytics.

7. Timetable and Management Plan

The team has developed a detailed timeline for each activity (see Table 6) involved in the project starting from 10/01/2026 to 09/30/2029. Each project year is divided into three columns, representing Fall, Spring and Summer semesters. The extra column before Year 1 represents summer 2026 when we aim to organize the kickoff meeting and set up the CI environment and project website before the project starts.

Table 6. Project Timetable

Tasks		Year 1			Year 2			Year 3		
		FA	SP	SU	FA	SP	SU	FA	SP	SU
Project Kickoff Meeting										
CI Environment Setup										
Core CI Skills Modules										
Urban Data Analytics Modules										
Urban Informatics Use-Cases Modules										
Webinar										
Summer School										
AAG Workshop										
PEARC Workshop										
ACSP Workshop										
UCGIS Workshop										
Integrate into VT & UTD Class										
Online Training Materials & Updates										
Evaluate & Refine Training Modules										

The project will start with a group kickoff meeting in Summer 2026 for all PIs and collaborators to discuss the strategies for implementing the CITY-AI courses, recruitment, management, and evaluation plan. The Core CI Skills Module will be implemented before the summer of the first year and we will finish the Urban Data Analytics modules and Urban Informatics Use-case modules by the end of the second year. We will set up the project website, Wiki page, and Slack channels by the beginning of Year 1 to promote our materials, build community, and gather early feedback. Webinars will take place during the spring and fall semesters of each year. We plan to organize a total of six workshops across the Annual Meeting of the AAG & ACSP, UCGIS Symposium, and PEARC. The workshops will feature two sessions: 1) workshop led by co-PI Michels focusing on how advanced CI works and how AI can be applied to urban informatics and urban management, targeting geography and urban planning communities who would benefit from learning the core CI skills and urban data analytics; and 2) workshop led by PI Lyu focusing on urban informatics research methods and use cases, targeting more technically oriented audiences in CI and computational science and showcasing how AI and CI can be used together for more effective urban management. All project team members will participate in the workshops either in person or remotely. The CITY-AI summer school will be held in Year 3 in Dallas, Texas. PIs Lyu, Michels, and Song, and their graduate students and research staff, will organize and deliver the summer school in person.

All CITY-AI course materials will be made openly available through the NSF-funded I-GUIDE Platform and the project website, and will be integrated into existing courses at Virginia Tech (e.g., GEOG 4424 Geospatial Data Science, GEOG 2984 Geospatial Tools, GEOG 2084 Principles of GIS) and the UTD (e.g., GISC 4381/6321 Spatial Data Science, CRIM 6332/GISC 6331 GIS Applications in Criminology). The PIs will collaboratively develop and select workshop and summer school content following an Outline–Implement–Evaluate–Refine process, with course guidelines, training materials, and instructional content refined iteratively based on team meetings and participant feedback.

8. Broader Impact

This project will broaden access to advanced CI- and AI-enabled methods for urban informatics and smart city research communities, while strengthening workforce development across disciplines including geography, urban planning, public policy, computer science, and social sciences. CITY-AI will expand participation in CI-enabled urban analytics by providing open training materials, hands-on workshops, and mentored learning experiences that lower technical barriers for students, educators, and early-career researchers, particularly from disciplines and institutions with historically limited engagement in advanced CI. The project will be scaled and sustained through partnerships with NSF-funded CI platforms (e.g., NSF ACCESS and I-GUIDE), professional organizations (e.g., AAG and UCGIS), and partner institutions. CITY-AI materials will be integrated into undergraduate and graduate curricula at Virginia Tech and the University of Texas at Dallas, ensuring sustained impact. By training a new generation of researchers and practitioners to use CI, AI, and real-time urban data, the project will enhance data-driven urban decision-making and support more resilient and equitable communities.

8. Broader Impact

F. Lyu and A. Michels are beginning investigators and have no prior NSF support as senior personnel.

X. Carol Song is PI for: (a) Grant: NSF OAC-2005632; \$31,497,100; 2020–2027. (b) Title: Category I: Anvil – A National Composable Advanced Computational Resource for the Future of Science and Engineering. (c) Results: Intellectual Merit: Established Anvil as a composable national HPC resource integrating large-scale clusters, GPUs, Kubernetes-based on-prem cloud, and a production software stack supporting dynamic configuration and cloud bursting. Broader Impacts: Lowered HPC access barriers via user platforms, REU programs, and workforce training; strengthened national competitiveness through startup engagement and innovation pathways. (d) Publications:

REFERENCE:

- [1] S. Wang, F. Lyu, S. Wang, C. E. Catlett, A. Padmanabhan, and K. Soltani, “Integrating CyberGIS and urban sensing for reproducible streaming analytics,” in *Urban Informatics*, Singapore: Springer Singapore, 2021, pp. 663–681.
- [2] G. S. Thakur, B. L. Bhaduri, J. O. Piburn, K. M. Sims, R. N. Stewart, and M. L. Urban, “PlanetSense,” in *Proceedings of the 23rd SIGSPATIAL International Conference on Advances in Geographic Information Systems*, Seattle Washington, 2015.
- [3] National Academies of Sciences, Engineering, and Medicine, Policy and Global Affairs, Division on Engineering and Physical Sciences, Division of Behavioral and Social Sciences and Education, Division on Earth and Life Studies, and Committee on Advancing a Systems Approach to Studying the Earth: A Strategy for the National Science Foundation, *Next generation earth systems science at the national science foundation*. Washington, D.C., DC: National Academies Press, 2022.
- [4] Panel on the Scope, Components, and Key Characteristics of a 21st-Century Data Infrastructure, Committee on National Statistics, Division of Behavioral and Social Sciences and Education, and National Academies of Sciences, Engineering, and Medicine, “Toward a 21st century national data infrastructure: Mobilizing information for the common good.” National Academies Press, Washington, D.C., 01-Mar-2023.

- [5] C. E. Catlett, P. H. Beckman, R. Sankaran, and K. K. Galvin, "Array of things: a scientific research instrument in the public way: platform design and early lessons learned," in *Proceedings of the 2nd international workshop on science of smart city operations and platforms engineering*, 2017, pp. 26–33.
- [6] M. Picone, S. Busanelli, M. Amoretti, F. Zanichelli, and G. Ferrari, *Advanced Technologies for Intelligent Transportation Systems*. Cham, Switzerland: Springer International Publishing, 2016.
- [7] Committee on Foundational Research Gaps and Future Directions for Digital Twins *et al.*, "Foundational Research Gaps and Future Directions for Digital Twins." National Academies Press, Washington, D.C., 28-Mar-2024.
- [8] S. Wang, "CyberGIS and spatial data science," *GeoJournal*, vol. 81, no. 6, pp. 965–968, Dec. 2016.
- [9] H. Xu *et al.*, "Smart mobility in the cloud: Enabling real-time situational awareness and cyber-physical control through a digital twin for traffic," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 3, pp. 3145–3156, Mar. 2023.
- [10] F. Lyu, L. Zhou, J. Park, F. Baig, and S. Wang, "Mapping Dynamic Human Sentiments of Heat Exposure with Location-Based Social Media Data," *International Journal of Geographical Information Science*, 2024.
- [11] F. Lyu, S. Wang, S. Y. Han, C. Catlett, and S. Wang, "An integrated cyberGIS and machine learning framework for fine-scale prediction of Urban Heat Island using satellite remote sensing and urban sensor network data," *Urban Informatics*, vol. 1, no. 1, p. 6, Sept. 2022.
- [12] Z. Lv, B. Hu, and H. Lv, "Infrastructure monitoring and operation for smart cities based on IoT system," *IEEE Trans. Industr. Inform.*, vol. 16, no. 3, pp. 1957–1962, Mar. 2020.
- [13] H. Zhang, R. Zhang, and J. Sun, "Developing real-time IoT-based public safety alert and emergency response systems," *Sci. Rep.*, vol. 15, no. 1, p. 29056, Aug. 2025.
- [14] J. Lee *et al.*, "An integrated approach to intelligent urban facilities management for real-time emergency response," *Autom. Constr.*, vol. 30, pp. 256–264, Mar. 2013.
- [15] T. W. Sanchez, H. Shumway, T. Gordner, and T. Lim, "The prospects of artificial intelligence in urban planning," *Int. J. Urban Sci.*, pp. 1–16, July 2022.
- [16] M. Ahmed, T. Rahim, and Z. Nasar, "Role of AI-Supported Serious Simulation Games in Urban Planning Education," in *2023 16th International Conference on Developments in eSystems Engineering (DeSE)*, IEEE, 2023, pp. 599–604.
- [17] J. P. White, A. Weeden, R. Deleon, T. Furlani, and M. D. Jones, "Overview of ACCESS allocated cyberinfrastructure usage," in *Practice and Experience in Advanced Research Computing 2024: Human Powered Computing*, Providence RI USA, 2024, pp. 1–10.
- [18] J. T. Palmer *et al.*, "Open XDMoD: A tool for the comprehensive management of high-performance computing resources," *Comput. Sci. Eng.*, vol. 17, no. 4, pp. 52–62, July 2015.
- [19] R. C. Vandewalle, W. C. Barley, A. Padmanabhan, D. S. Katz, and S. Wang, "Understanding the multifaceted geospatial software ecosystem: a survey approach," *Geogr. Inf. Syst.*, vol. 35, no. 11, pp. 2168–2186, Nov. 2021.
- [20] F. Lyu, "Converging GeoAI and CyberGIS for human geography," in *GeoAI and Human Geography*, Cham: Springer Nature Switzerland, 2025, pp. 107–121.
- [21] D. Croushore, "Frontiers of real-time data analysis," *J. Econ. Lit.*, vol. 49, no. 1, pp. 72–100, Mar. 2011.
- [22] S. Gao, "Geospatial artificial intelligence (GeoAI)," *Geography*. Oxford University Press, 24-Mar-2021.
- [23] X. Huang and J.-Y. Kang, "High-performance geospatial computing and computationally reproducible human geography," in *GeoAI and Human Geography*, Cham: Springer Nature Switzerland, 2025, pp. 123–132.
- [24] S. Wang, *I-GUIDE platform for geospatial data-intensive convergence research and education*, in *Practice and Experience in Advanced Research Computing 2025: The Power of Collaboration*. New York, NY, USA: ACM, 2025, pp. 1–5.
- [25] F. Caprotti *et al.*, "Why does urban Artificial Intelligence (AI) matter for urban studies? Developing

- research directions in urban AI research,” *Urban Geogr.*, vol. 45, no. 5, pp. 883–894, May 2024.
- [26] X. Ye *et al.*, “Artificial intelligence in urban science: why does it matter?,” *Ann. GIS*, vol. 31, no. 2, pp. 181–189, June 2025.
- [27] A. Luusua, J. Ylipulli, M. Foth, and A. Aurigi, “Urban AI: understanding the emerging role of artificial intelligence in smart cities,” *AI Soc.*, vol. 38, no. 3, pp. 1039–1044, 2023.
- [28] A. Michels, J. Park, J.-Y. Kang, and S. Wang, “An Areal approach to spatial accessibility analysis,” *Geogr. Anal.*, vol. 57, no. 2, pp. 233–269, Apr. 2025.
- [29] A. C. Michels, J. Park, J.-Y. Kang, and S. Wang, “SPASTC: A spatial partitioning algorithm for scalable travel-time computation,” *Geogr. Inf. Syst.*, vol. 38, no. 5, pp. 803–824, May 2024.
- [30] K. Thomas, “Jupyter Notebooks - a publishing format for reproducible computational workflows,” in *Positioning and Power in Academic Publishing: Players, Agents and Agendas*, IOS Press, 2016.
- [31] A. Michels, A. Padmanabhan, Z. Li, and S. Wang, “EasyScienceGateway: A new framework for providing reproducible user environments on science gateways,” *Computation and Concurrency: Practice and Experience*, vol. 36, Oct. 2023.
- [32] D. Sinton, “GIS&T body of knowledge,” <https://www.ucgis.org/gis-t-body-of-knowledge>, 2025. .
- [33] I-GUIDE, “Convergence Curriculum for Geospatial Data Science,” <https://i-guide.io/projects/convergence-curriculum-for-geospatial-data-science/>, 2026. .
- [34] E. Shook, F. Bowlick, A. Padmanabhan, and K. Kemp, “The hour of cyberinfrastructure (hour of CI): Early findings from pilot study to build cyber literacy for GIScience,” in *2021 IEEE/ACM Ninth Workshop on Education for High Performance Computing (EduHPC)*, St. Louis, MO, USA, 2021, pp. 9–15.
- [35] F. Lyu, D. Kiv, X. Ma, and S. Wang, “A video machine learning framework for spatiotemporal analysis of complex urban dynamics,” *Trans. GIS*, vol. 29, no. 5, Aug. 2025.
- [36] X. Ma, Y. Song, F. Lyu, Y. Yang, X. Li, and S. Zhong, “Revitalizing Cities: The 5R Framework Approach to Urban Retrofitting and Big Data Insights,” *Growth and*, 2025.
- [37] X. Li, X. Ma, F. Lyu, and Y. Song, “Examine the environmental inequity impact of urban heat mitigation on redlining legacy: case study of Charlotte’s retrofitting, 2001–2020,” *Front. Environ. Sci.*, vol. 11, Sept. 2023.
- [38] F. Lyu, Z. Yang, C. Diao, and S. Wang, “Multistream STGAN: A Spatiotemporal Image Fusion Model With Improved Temporal Transferability,” *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, vol. 18, pp. 1562–1576, 2025.
- [39] S. Wu *et al.*, “Spatiotemporal assessments of nutrients and water quality in coastal areas using remote sensing and a spatiotemporal deep learning model,” *Int. J. Appl. Earth Obs. Geoinf.*, vol. 112, no. 102897, p. 102897, Aug. 2022.
- [40] C. Wang *et al.*, “A deep trajectory clustering method based on sequence-to-sequence autoencoder model,” *Trans. GIS*, vol. 26, no. 4, pp. 1801–1820, June 2022.
- [41] Y. Kang, F. Lyu, and S. Wang, “NetPointLib: Library for large-scale spatial network point data fusion and analysis,” in *Practice and Experience in Advanced Research Computing 2024: Human Powered Computing*, Providence RI USA, 2024, pp. 1–4.
- [42] F. Lyu, Z. Yang, Z. Xiao, C. Diao, and J. Park, “CyberGIS for scalable remote sensing data fusion,” *Practice and Experience*, 2022.
- [43] F. Lyu, Z. Xu, X. Ma, S. Wang, Z. Li, and S. Wang, “A vector-based method for drainage network analysis based on LiDAR data,” *Comput. Geosci.*, vol. 156, no. 104892, p. 104892, Nov. 2021.
- [44] A. Michels, M. Kotak, A. Padmanabhan, J. Speaks, and S. Wang, “Providing accessible software environments across science gateways and HPC,” in *Practice and Experience in Advanced Research Computing 2024: Human Powered Computing*, Providence RI USA, 2024, pp. 1–4.
- [45] A. C. Michels, A. Padmanabhan, Z. Xiao, M. Kotak, F. Baig, and S. Wang, “CyberGIS-Compute: Middleware for democratizing scalable geocomputation,” *SoftwareX*, vol. 26, no. 101691, p. 101691, May 2024.
- [46] S. Y. Han *et al.*, “CyberGIS-Vis for Democratizing Access to Scalable Spatiotemporal Geovisual Analytics: A Case Study of COVID-19,” in *Proceedings of the 5th ACM SIGSPATIAL International*

- Workshop on Spatial Computing for Epidemiology*, Atlanta GA USA, 2024, pp. 19–22.
- [47] J.-Y. Kang *et al.*, “Rapidly measuring spatial accessibility of COVID-19 healthcare resources: a case study of Illinois, USA,” *Int. J. Health Geogr.*, vol. 19, no. 1, p. 36, Sept. 2020.
 - [48] J.-Y. Kang, J. Aldstadt, A. Michels, R. Vandewalle, and S. Wang, “CyberGIS-Jupyter for spatially explicit agent-based modeling,” in *Proceedings of the 2nd ACM SIGSPATIAL International Workshop on GeoSpatial Simulation*, Chicago Illinois, 2019.
 - [49] J.-Y. Kang, B. F. Farkhad, M.-P. S. Chan, A. Michels, D. Albarracin, and S. Wang, “Spatial accessibility to HIV testing, treatment, and prevention services in Illinois and Chicago, USA,” *PLoS One*, vol. 17, no. 7, p. e0270404, July 2022.